

Supplemental Material: Yukawa screening derivation of the bond-valence rule

Michael L. Whittaker,^{1,2} Pan Wang,¹ Chunhui Li,³ Naman Katyal,¹ and Piotr Zarzycki²

¹*Energy Geosciences Division, Lawrence Berkeley National Laboratory, Berkeley, CA 94720*

²*Materials Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, CA 94720*

³*Meta Platforms, Inc., Menlo Park, CA 94025*

DATA PROVENANCE

The figures and tables in this Supplemental Material cover the same 103 cation–oxygen species analyzed in the main text, spanning *s*-, *p*-, *d*-, and *f*-block cations.

Source structures are DFT-relaxed entries from the Materials Project. For each structure, bond strengths were obtained by solving the valence-sum and loop constraints on the cation–oxygen bond graph, and (R_0, B) were determined by linear least squares. For rank-deficient bond graphs (fewer independent bonds than parameters), an anion-centered reparameterization was used. Any remaining ill-conditioned cases were resolved with the constrained nonlinear procedure of Ref. 1, with R_0 bounds $(-10, 10)$ Å. Coordination numbers were determined from the bond graph.

The DFT-centroid comparison in Fig. 2 uses ten *s*-block oxides in panel a (Li_2O , Na_2O , K_2O , Rb_2O , Cs_2O , BeO , MgO , CaO , SrO , and BaO) and 25 non-*s*-block oxides in panel b: Al_2O_3 , SiO_2 , GeO_2 , SnO_2 , PbO , TiO_2 , V_2O_5 , CoO , NiO , CuO , ZnO , ZrO_2 , MoO_3 , PdO , Ag_2O , HfO_2 , Ta_2O_5 , Au_2O_3 , La_2O_3 , Gd_2O_3 , Yb_2O_3 , B_2O_3 , P_2O_5 , As_2O_3 , and Sb_2O_3 . The panel-*b* regression is defined a priori as a test of the Yukawa screening picture only for ground-state binary oxides whose cation environments approximate an isotropic first-shell ionic screening cloud. It therefore uses the 21 filled-symbol oxides and excludes B_2O_3 , P_2O_5 , As_2O_3 , and Au_2O_3 , whose ground-state structures are dominated by directional, non-isotropic bonding motifs. Those four compounds are shown as open symbols. The 21 oxides shown as filled symbols in Fig. 2(b) are the 25 listed above minus the four directional cases just named. No further data-driven exclusion was applied beyond this fixed list. The excluded oxides are the cases in this benchmark whose ground-state cation environments are not compact first-shell ionic polyhedra but directional or terminal-oxyanion motifs, so a single shell centroid m_i is not an adequate descriptor of the screening cloud.

COORDINATION-NUMBER-RESOLVED B - R_0 FIT LINES

Figures 1–15 show the n -resolved B vs. R_0 fit lines for every species in the 103-cation set, grouped by block: Group 1 (Fig. 1), Group 2 (Fig. 2), $3d/4d/5d$ transition metals (Figs. 3–10), *p*-block (Figs. 11–13), and *f*-block lanthanides plus actinides (Figs. 14–15). Each caption quotes the species panels of its page in reading order, either in full or as a first–last range. Within each block, panels are ordered alphabetically by element symbol and oxidation state. Within each coordination shell, the fitted parameters lie on a linear family $B = \beta^{(n)} R_0 + \beta_0^{(n)}$, and the lines converge toward cation-specific characteristic pairs (R_0^*, B^*) as described in the main text. Together, these atlas pages display every coordination-resolved fit that contributes to the pole form $\beta = 1/\ln(z/n)$ shown in Fig. 1 of the main text.

Line-fitting procedure. Each DFT-relaxed structure contributes one independently fitted (R_0, B) pair. With the solved S_{ij} as data, the model $R_{ij} = R_0 - B \ln S_{ij}$ is linear in both parameters, which are determined jointly by ordinary least squares (OLS) over the structure’s cation–oxygen bonds. R_0 is fixed and B re-optimized only in the rare bound-projection fallback, when the unconstrained R_0 falls outside the pipeline’s R_0 bounds. A structure enters coordination class n only when its cation sites share an unambiguous oxygen coordination number in the bond graph. Mixed-coordination structures are left unbinned. For each species and coordination class with at least five structures, the line $B = \beta^{(n)} R_0 + \beta_0^{(n)}$ is fitted both by OLS and by random sample consensus (RANSAC), with residual threshold equal to twice the median absolute OLS residual and floored at 0.01 Å, and the RANSAC line is used whenever available. The reported slope and intercept come from an OLS refit on the RANSAC inlier set, while their standard errors and confidence intervals come from a 400-resample bootstrap over that inlier set. The population N_n of a line is the number of inlier structures in this final refit. It supplies the weights $w = \max(N_n, 1)$ used in the goodness-of-fit metrics of main-text Fig. 1 and in the characteristic-pair intersections below. Lines enter Fig. 1 of the main text only if they retain at least four inliers, have slope standard error at most 5, and have $R^2 \geq 0.3$.

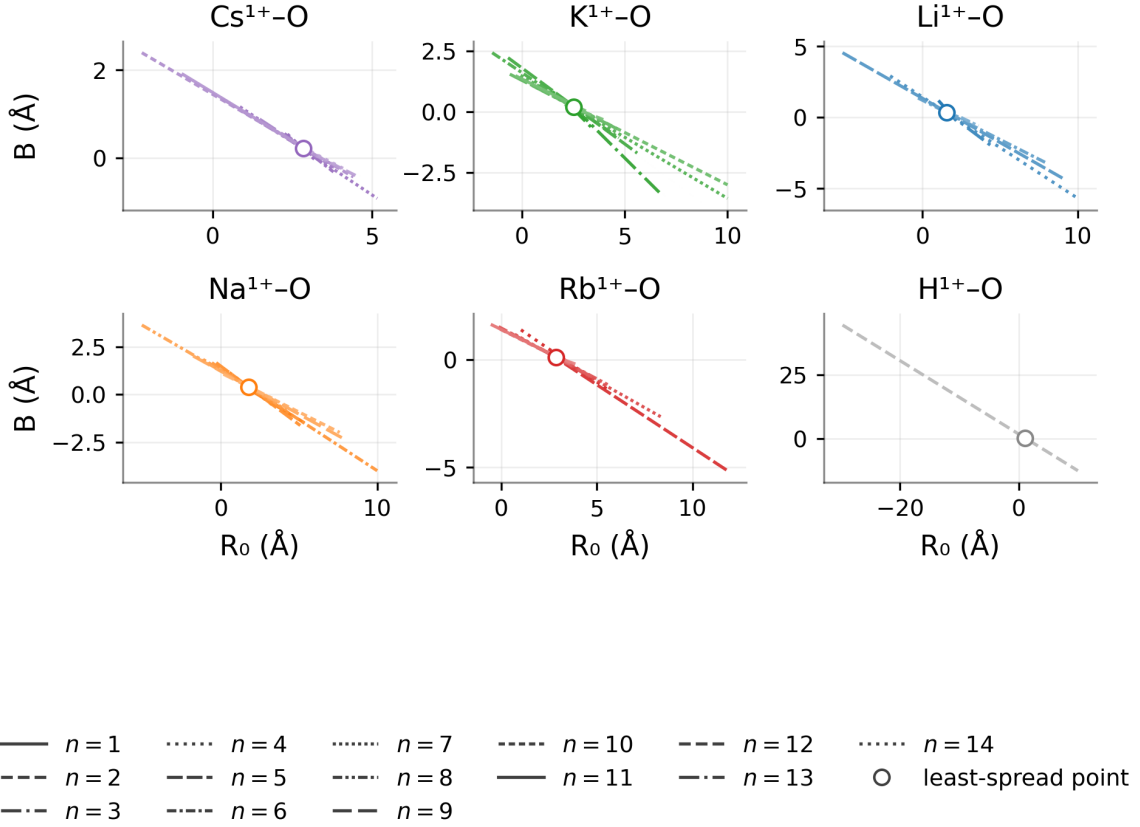


FIG. 1. n -resolved B vs. R_0 fit lines for Group 1 X-O pairs (Cs, K, Li, Na, Rb, H). Line styles distinguish coordination numbers n (legend), and the open circle marks the least-spread intersection point. The convergence of the lines defines the characteristic pair (R_0^*, B^*) for each cation.

Weighting scheme for the goodness-of-fit metrics. The weighted R^2 and mean absolute error reported with Fig. 1 of the main text use weights $w_i = \max(N_i, 1)$, where N_i is the number of DFT-relaxed structures contributing to the linear B - R_0 fit at species i and coordination n . This gives more populous (species, n) cells more leverage, in proportion to the number of independent structures backing each fitted slope β_i . Letting $\hat{\beta}_i = 1/\ln(z_i/n)$ be the parameter-free closed-form prediction and $\bar{\beta}_w = \sum_i w_i \beta_i / \sum_i w_i$, the reported metrics are

$$R_w^2 = 1 - \frac{\sum_i w_i (\beta_i - \hat{\beta}_i)^2}{\sum_i w_i (\beta_i - \bar{\beta}_w)^2}, \quad \text{MAE}_w = \frac{\sum_i w_i |\beta_i - \hat{\beta}_i|}{\sum_i w_i}.$$

Pooling 68 species at $n = 4$ ($\sum w = 13,370$) with 82 species at $n = 6$ ($\sum w = 17,162$) – the $z \neq n$ subset shown in Fig. 1 – gives $R_w^2 = 0.986$ and $\text{MAE}_w = 0.15$ for the parameter-free closed form $\beta = 1/\ln(z/n)$. The $n = 4$ panel alone gives $R_w^2 = 0.973$ with $\text{MAE}_w = 0.23$, and the $n = 6$ panel gives $R_w^2 = 0.979$ with $\text{MAE}_w = 0.089$.

Exclusion of $z = n$ species from Fig. 1. Species at $z = n$ (U^{4+} , Mn^{4+} , Ti^{4+} , V^{4+} , Zr^{4+} , Ir^{4+} , P^{4+} , Pb^{4+} , and Te^{4+} for $n = 4$, and Mo^{6+} and W^{6+} for $n = 6$) are omitted from Fig. 1 of the main text. At $z = n$ the valence-sum rule fixes the mean bond valence at unity but not each S_{ij} individually. For a degenerate shell with equal bond lengths the rule is satisfied for any B with $R_0 = \bar{R}$, leaving the B - R_0 slope β formally unconstrained. For nearly degenerate shells the S_{ij} differ and B is finite, but the fit is ill-conditioned because the intercept ceases to constrain the softness at $z = n$ (see the screening-collapse discussion in the main text). Accordingly, the fits exhibit very large variance (see the $z = n$ panels in Figs. 3–10 and 11–15, where the coordination-resolved lines have widely scattered slopes and intercepts). The closed-form expression $\beta = 1/\ln(z/n)$ correctly diverges at $z = n$, and the divergence itself is a quantitative prediction of the Yukawa screening picture. We plot only $z \neq n$ species so that finite, well-conditioned fitted slopes are not visually conflated with the formal pole. Fig. 1 therefore shows the 94 species (150 fitted slopes) for which $z \neq n$ at $n = 4$ or $n = 6$. Six species (Cl^{1-} , C^{4+} , Hg^{1+} , H^{1+} , N^{3+} , S^{4+}) lack a fittable line at either $n = 4$ or $n = 6$, P^{4+} lies exactly at the $n = 4$ pole, and Te^{6+} and U^{6+} fall below the $R^2 \geq 0.3$ acceptance threshold at $n = 6$.

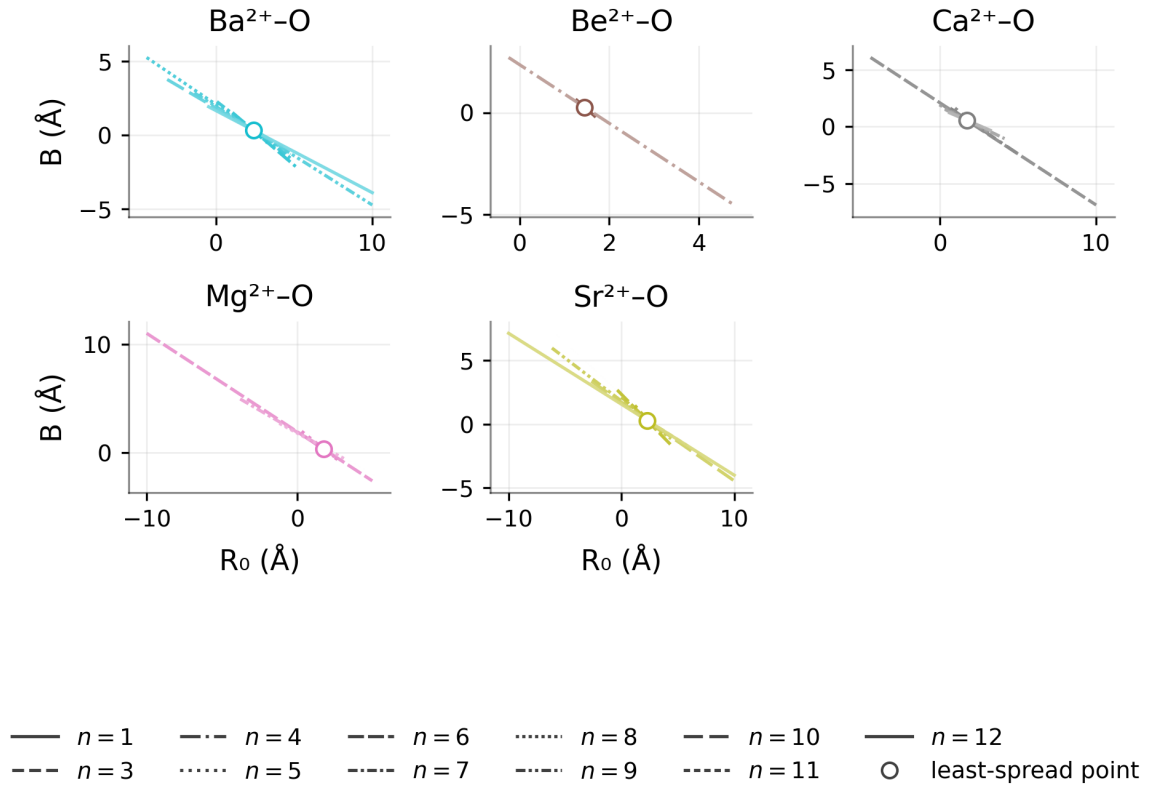


FIG. 2. n -resolved B vs. R_0 fit lines for Group 2 X-O pairs (Ba, Be, Ca, Mg, Sr).

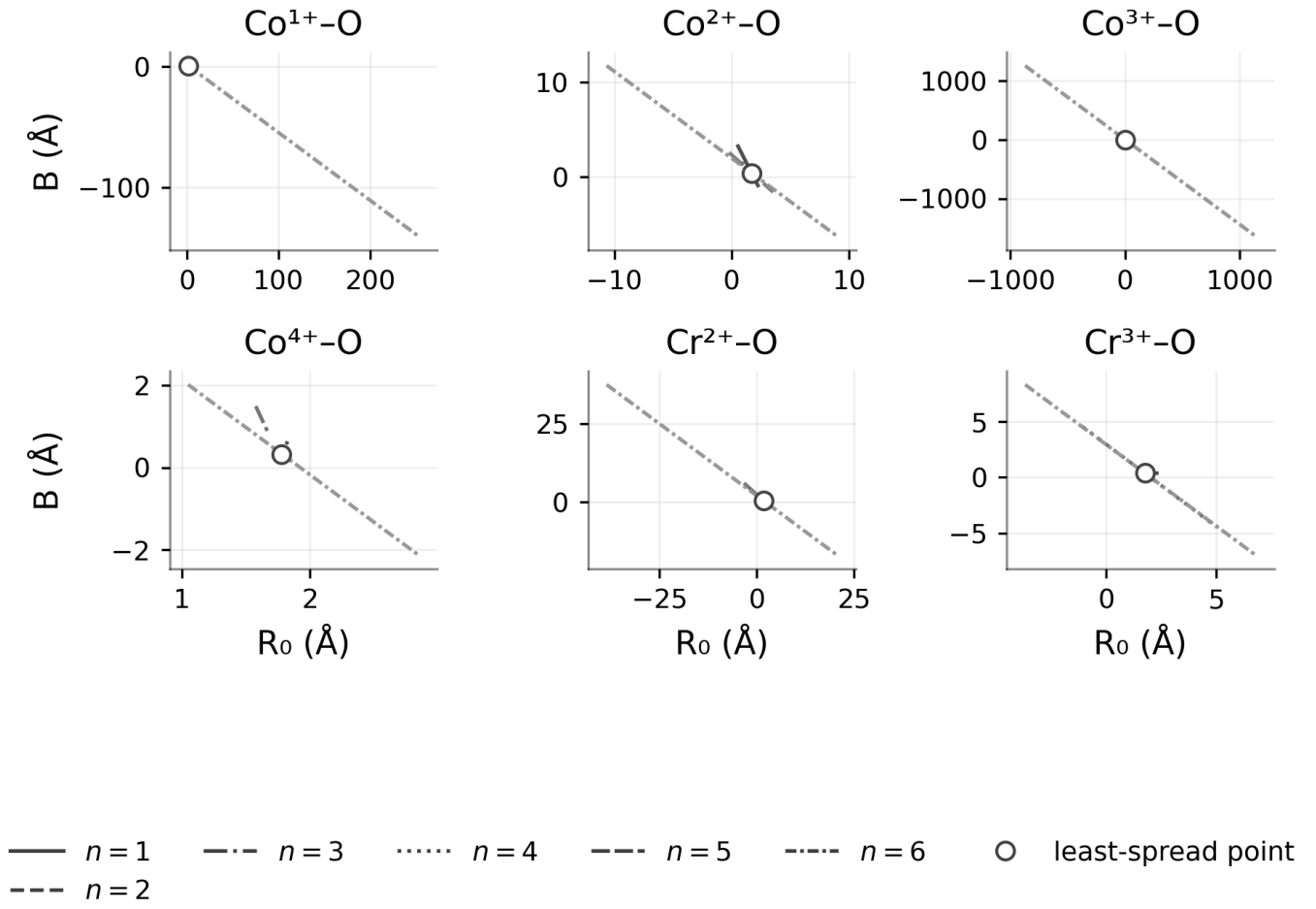
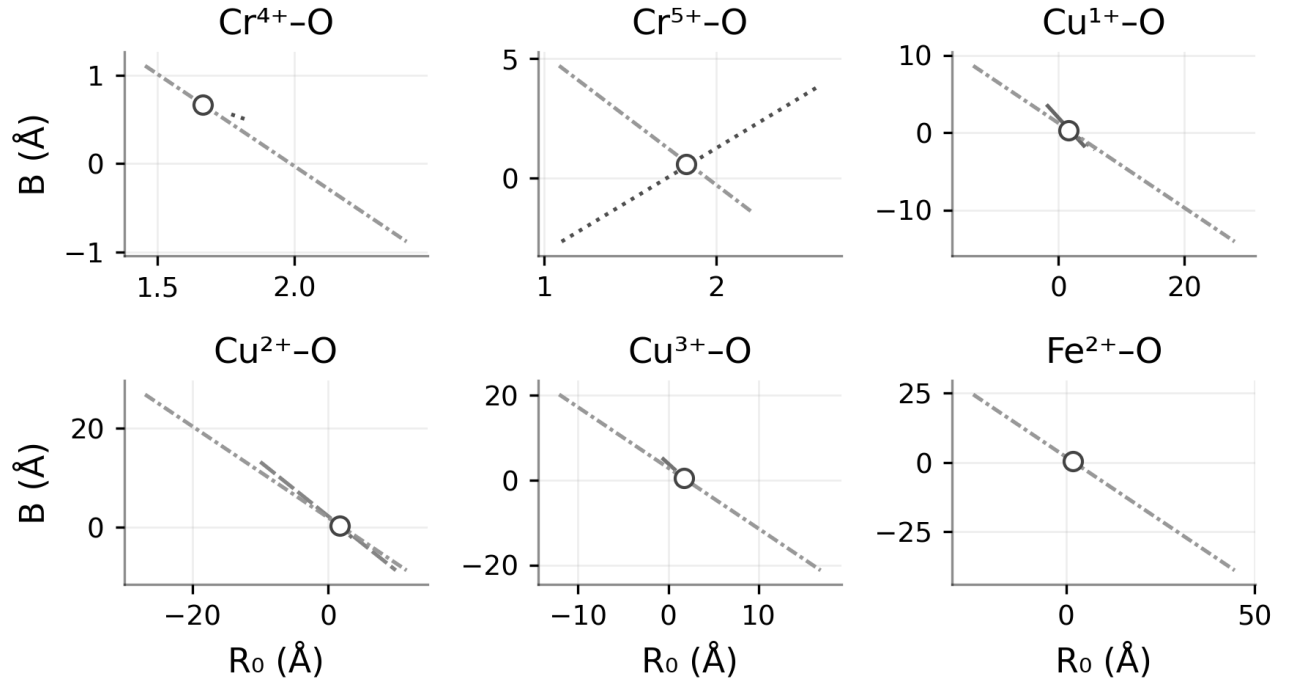


FIG. 3. n -resolved B vs. R_0 for oxidation-state-resolved $3d$ X-O pairs (Co¹⁺, Co²⁺, Co³⁺, Co⁴⁺, Cr²⁺, Cr³⁺).



— $n=1$ -.- $n=3$ $n=4$ --- $n=5$ -.-.- $n=6$ ○ least-spread point
 - - - $n=2$

FIG. 4. n -resolved B vs. R_0 for oxidation-state-resolved 3d X-O pairs (Cr^{4+} , Cr^{5+} , Cu^{1+} , Cu^{2+} , Cu^{3+} , Fe^{2+}).

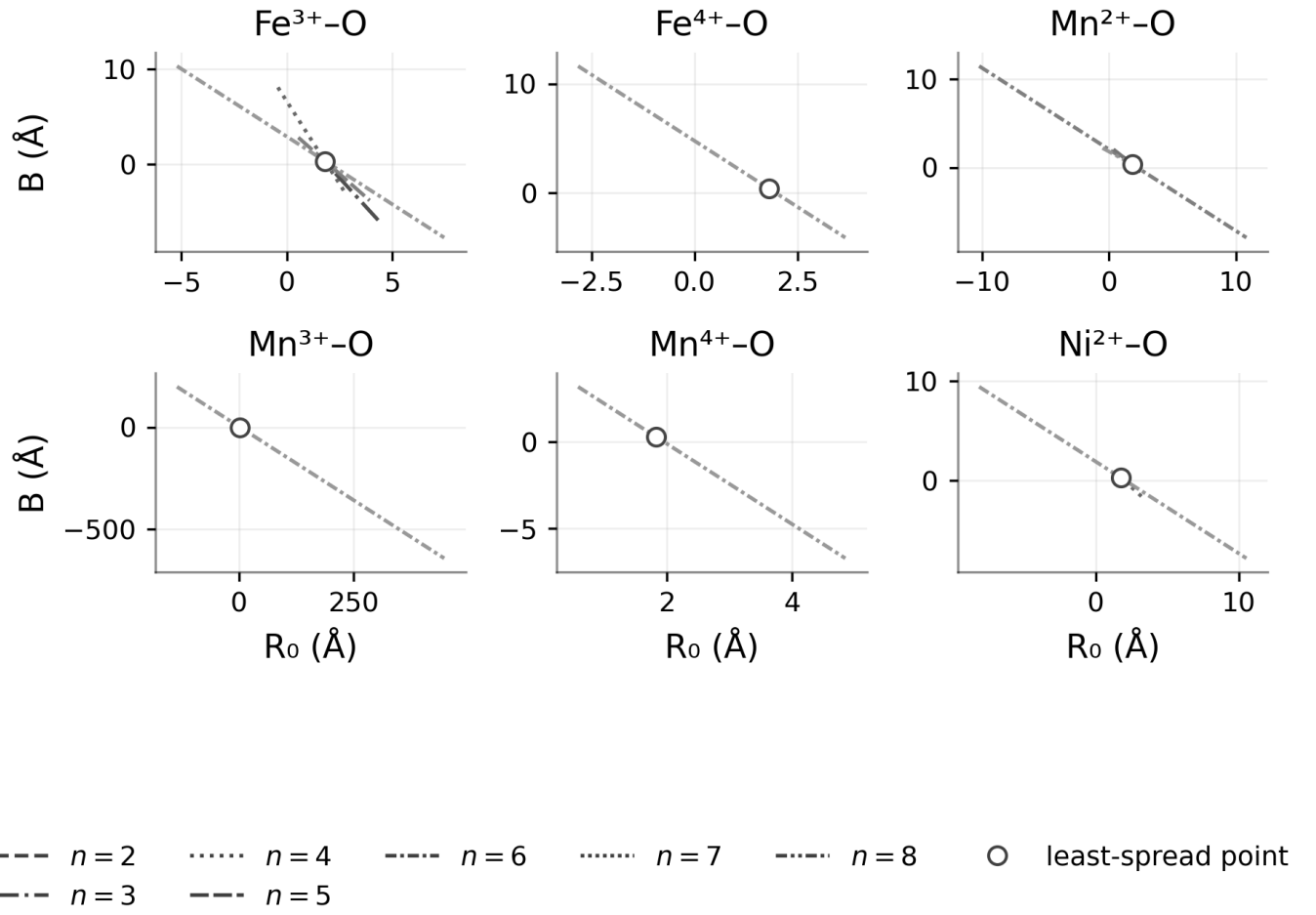


FIG. 5. n -resolved B vs. R_0 for oxidation-state-resolved $3d$ X-O pairs (Fe^{3+} , Fe^{4+} , Mn^{2+} , Mn^{3+} , Mn^{4+} , Ni^{2+}).

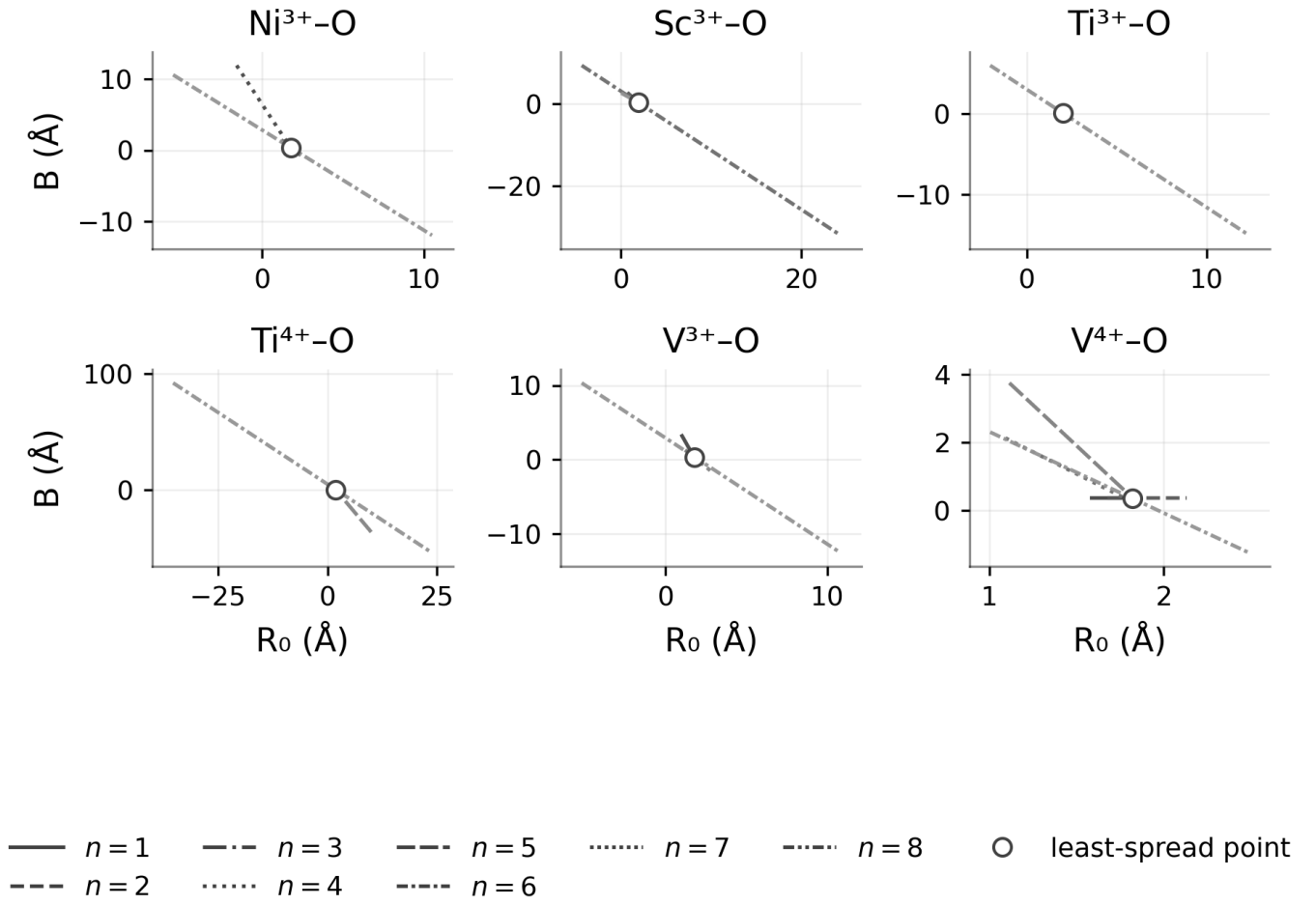


FIG. 6. n -resolved B vs. R_0 for oxidation-state-resolved 3d X-O pairs (Ni^{3+} , Sc^{3+} , Ti^{3+} , Ti^{4+} , V^{3+} , V^{4+}).

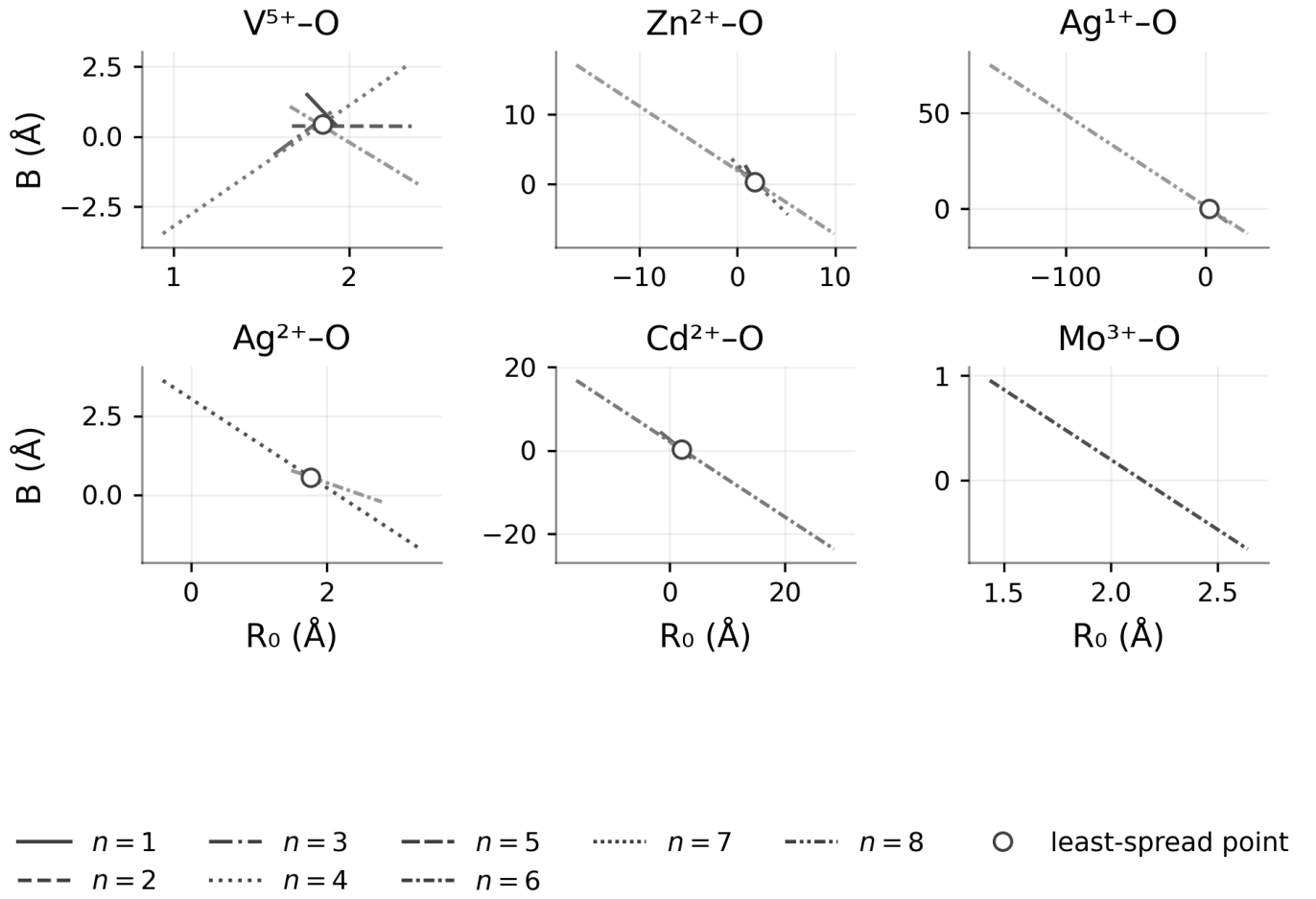


FIG. 7. n -resolved B vs. R_0 for oxidation-state-resolved 3d/4d X-O pairs (V^{5+} , Zn^{2+} , Ag^{1+} , Ag^{2+} , Cd^{2+} , Mo^{3+}).

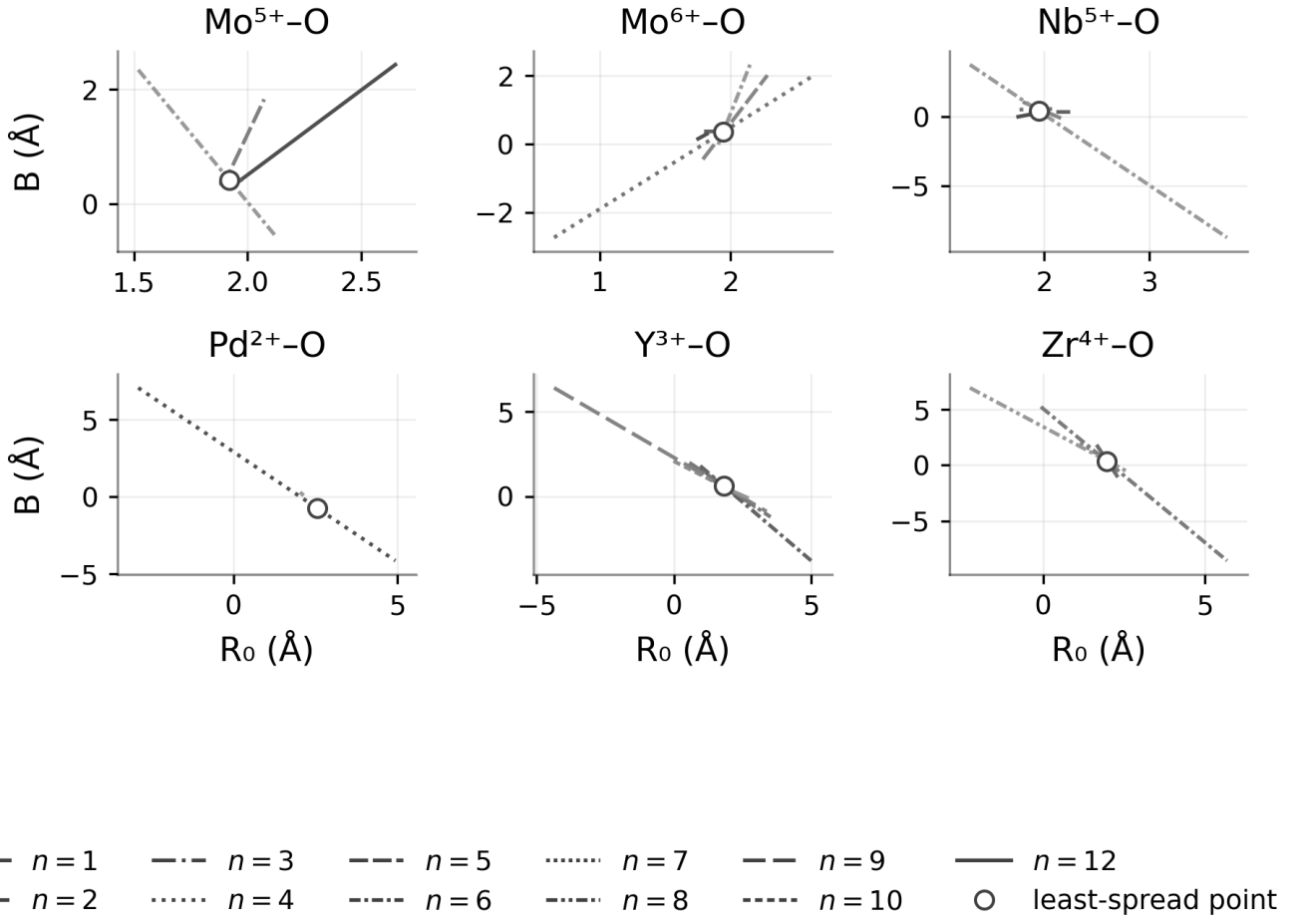


FIG. 8. n -resolved B vs. R_0 for oxidation-state-resolved $4d$ X-O pairs (Mo^{5+} , Mo^{6+} , Nb^{5+} , Pd^{2+} , Y^{3+} , Zr^{4+}).

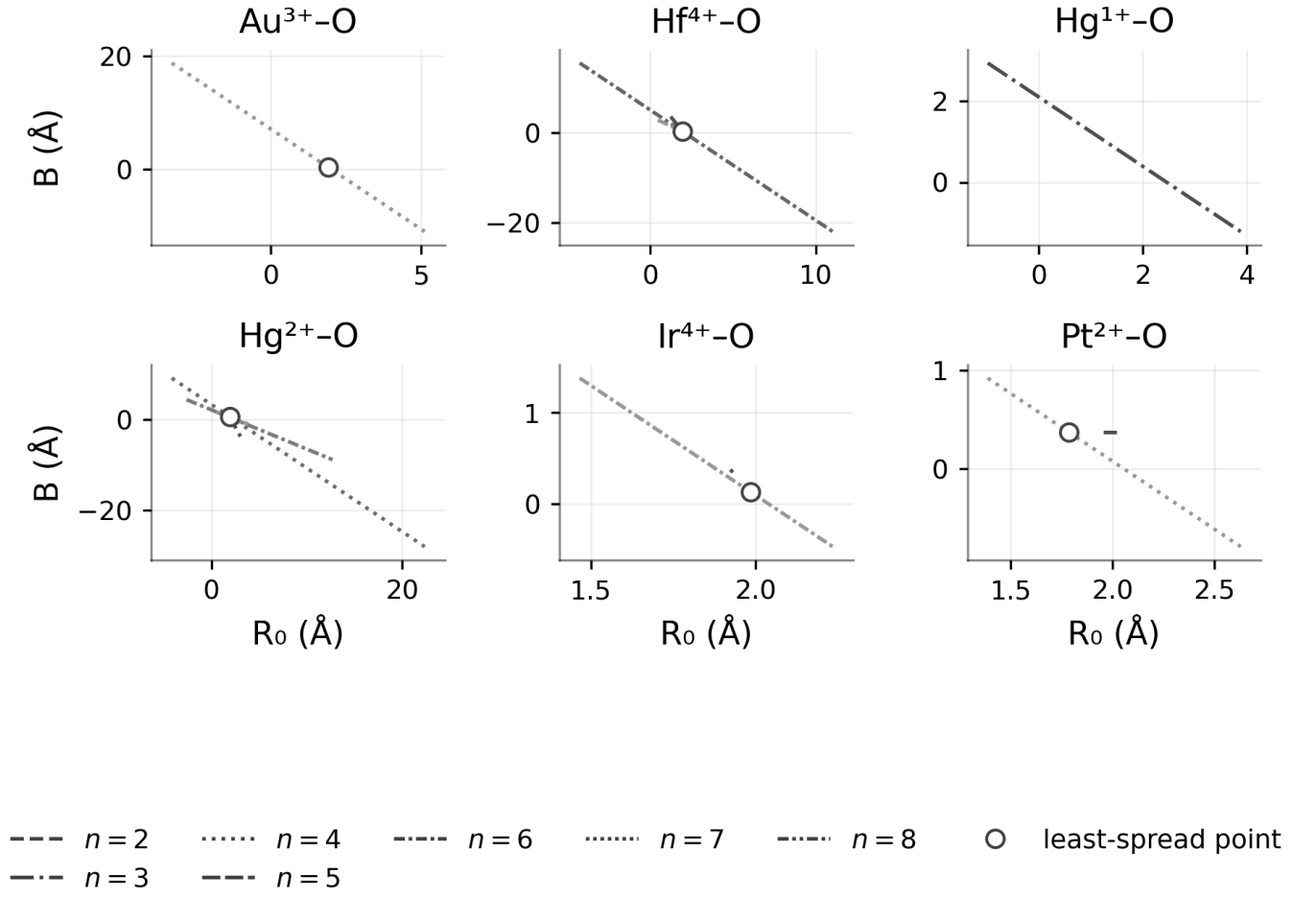


FIG. 9. n -resolved B vs. R_0 for oxidation-state-resolved $5d$ X-O pairs (Au^{3+} , Hf^{4+} , Hg^{1+} , Hg^{2+} , Ir^{4+} , Pt^{2+}).

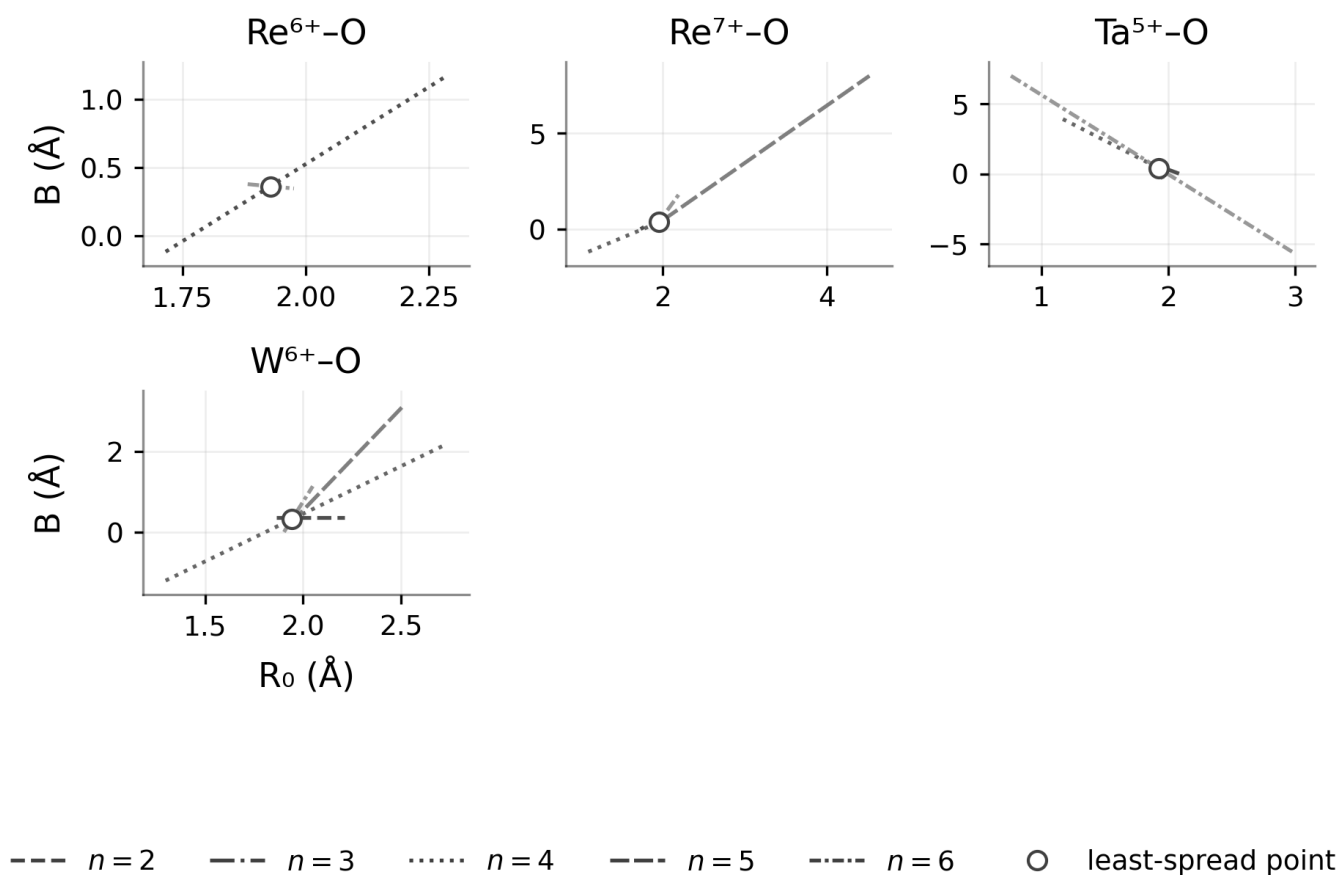


FIG. 10. n -resolved B vs. R_0 for oxidation-state-resolved $5d$ X-O pairs (Re^{6+} , Re^{7+} , Ta^{5+} , W^{6+}).

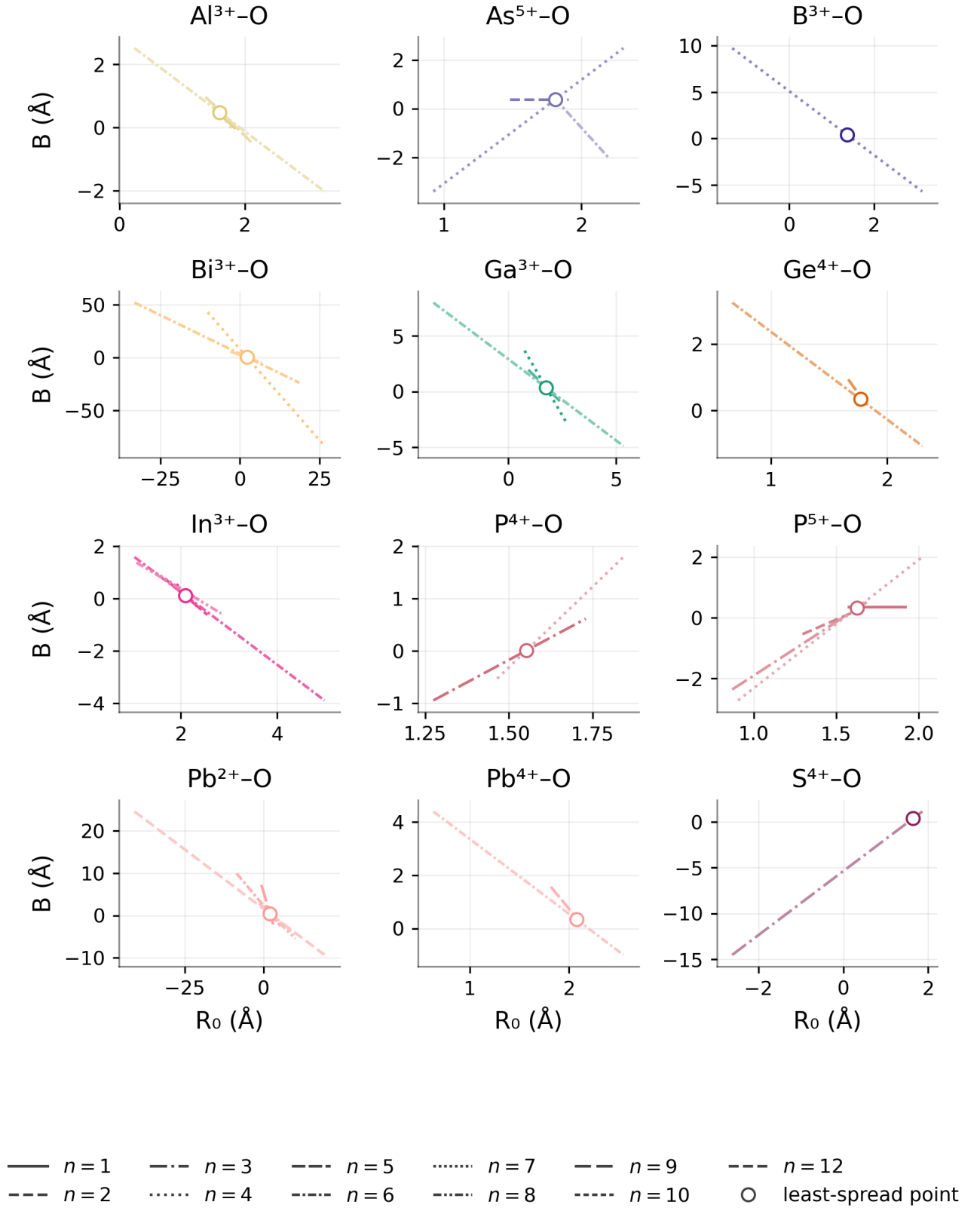


FIG. 11. n -resolved B vs. R_0 for oxidation-state-resolved p -block X-O pairs ($\text{Al}^{3+}\text{-S}^{4+}$).

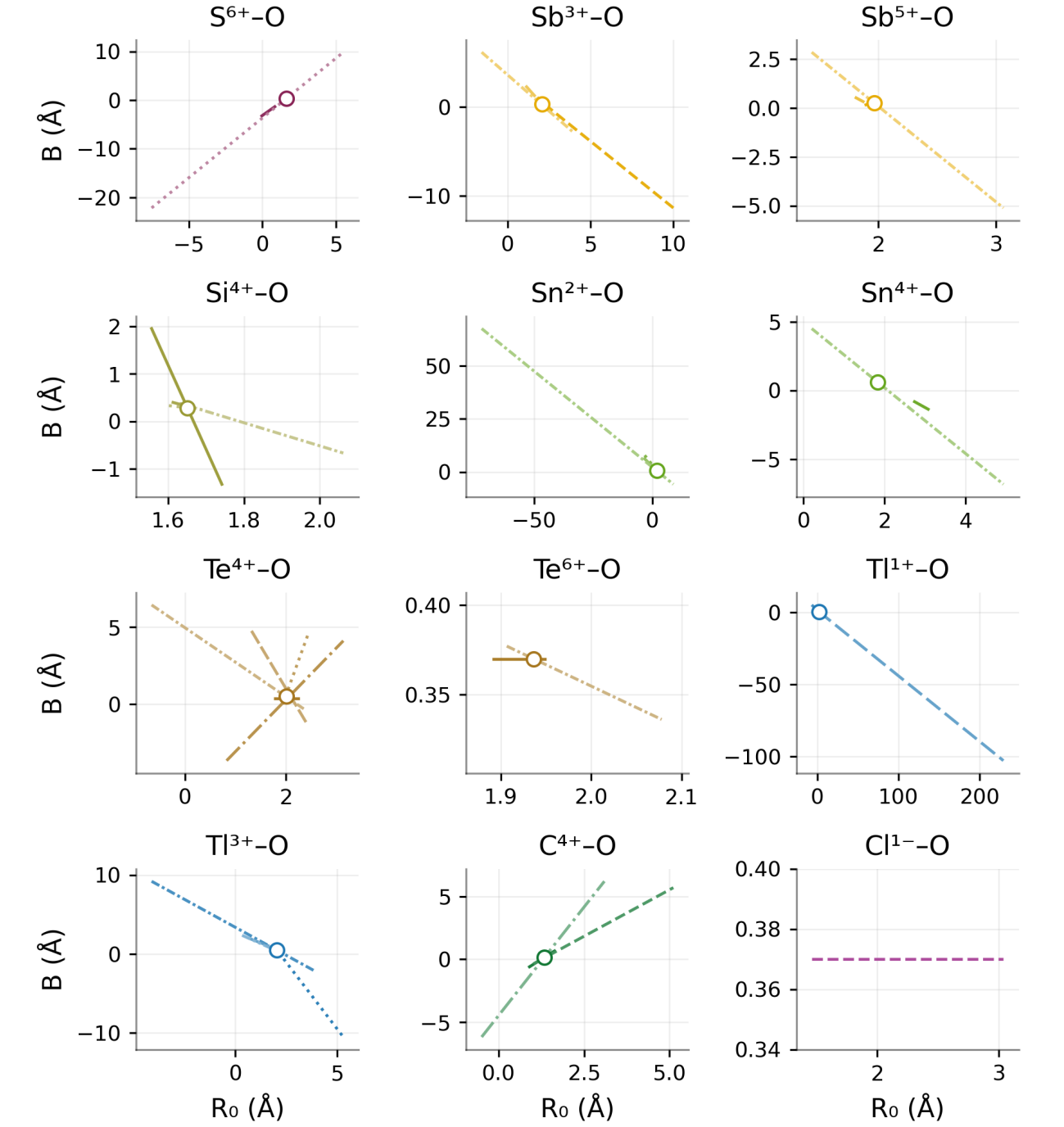


FIG. 12. n -resolved B vs. R_0 for oxidation-state-resolved p -block X-O pairs (S^{6+} , Sb^{3+} , Sb^{5+} , Si^{4+} , Sn^{2+} , Sn^{4+} , Te^{4+} , Te^{6+} , Tl^{1+} , Tl^{3+} , C^{4+} , Cl^{1-}).

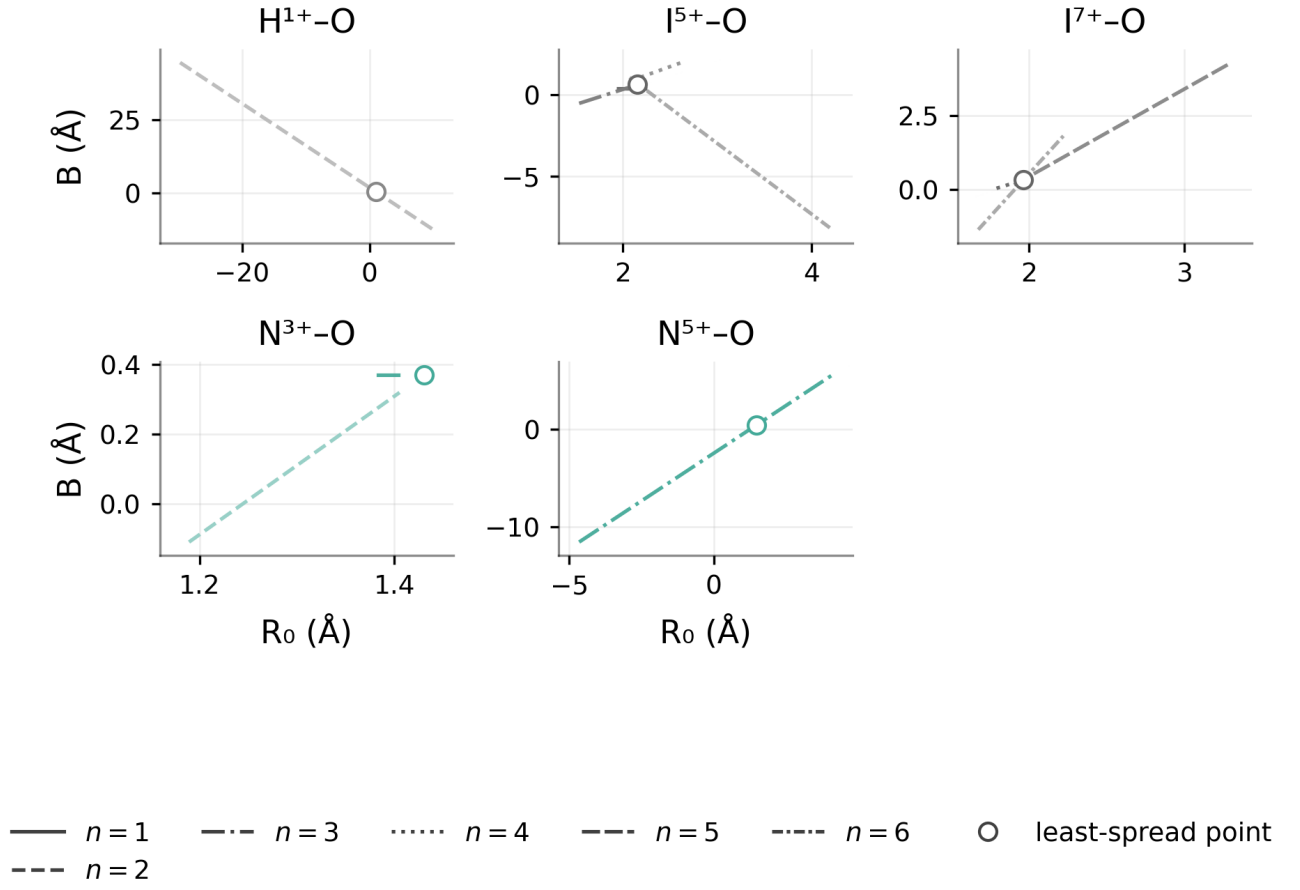


FIG. 13. n -resolved B vs. R_0 for the remaining p -block and hydrogen X-O pairs (H^{1+} , I^{5+} , I^{7+} , N^{3+} , N^{5+}).

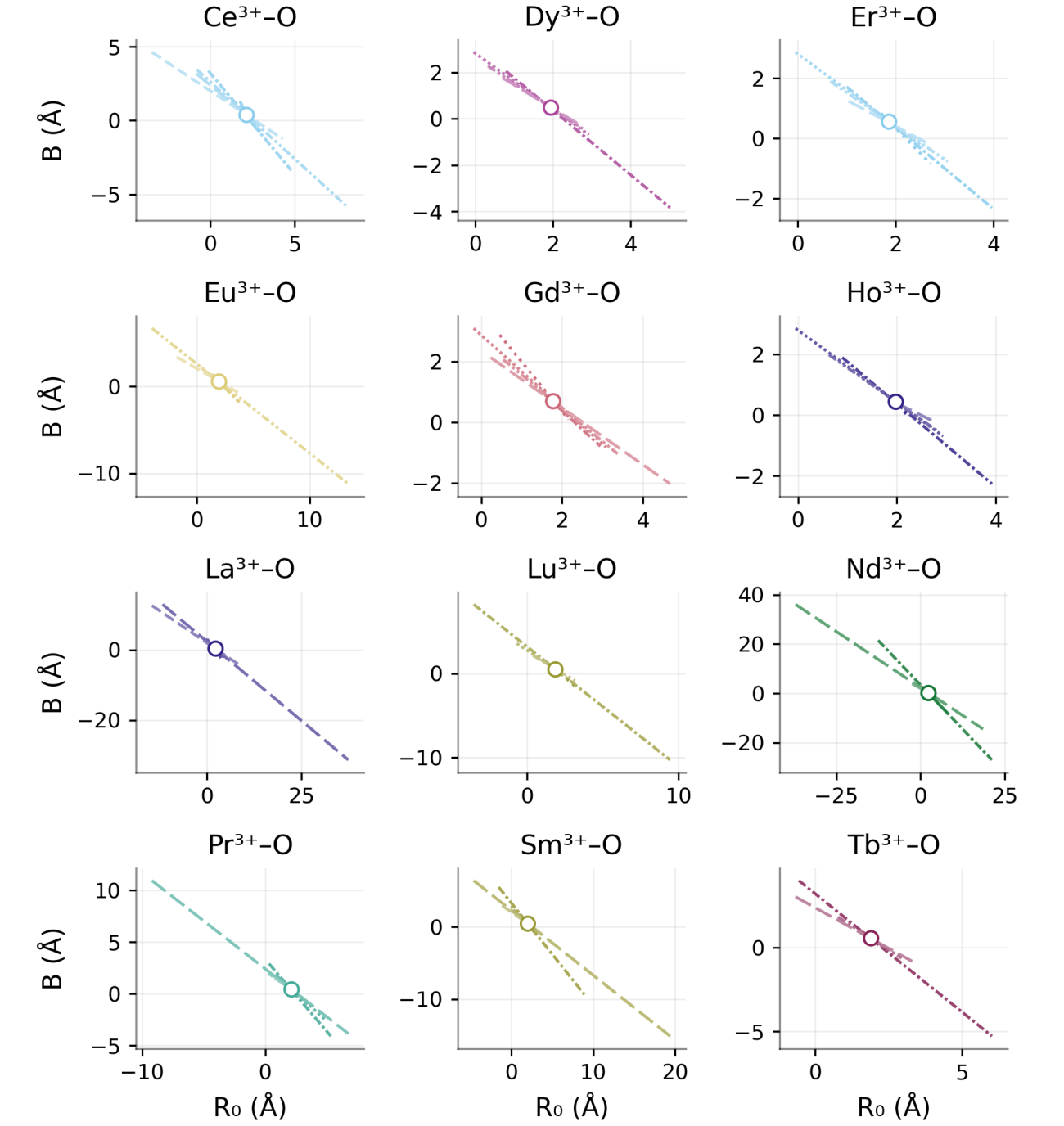
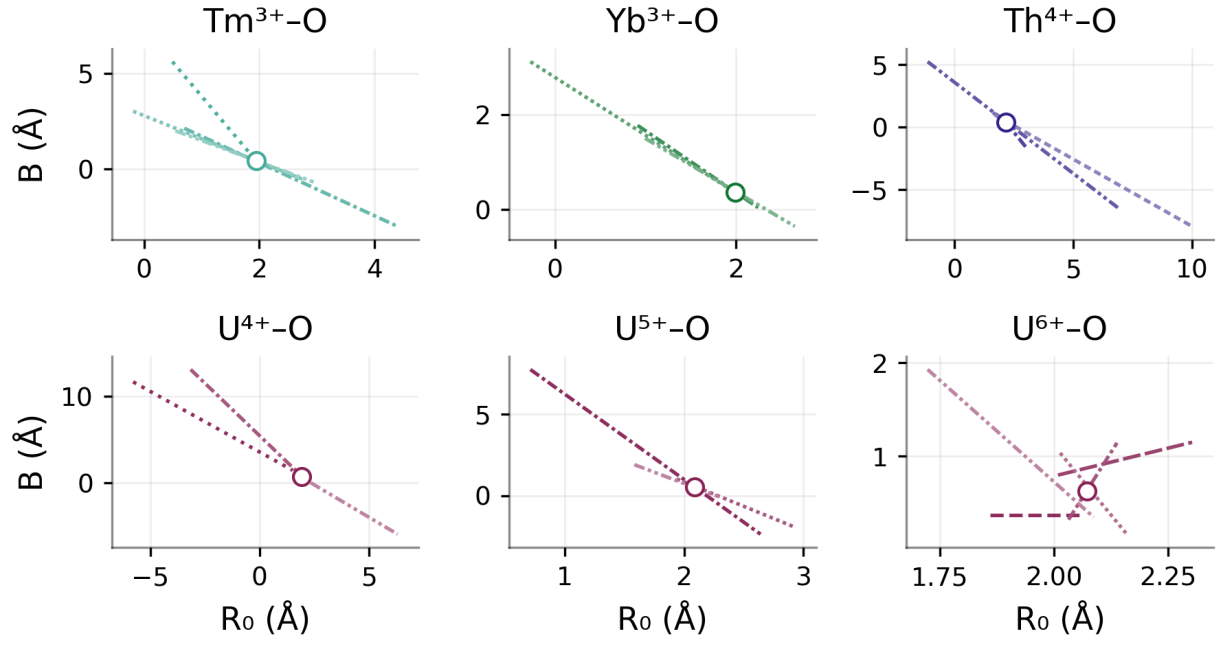


FIG. 14. n -resolved B vs. R_0 for f -block trivalent lanthanide X-O pairs (Ce³⁺-Tb³⁺).



--- $n=2$ --- $n=5$ $n=7$ --- $n=9$ $n=10$ ○ least-spread point
 $n=4$ $n=6$ $n=8$

FIG. 15. n -resolved B vs. R_0 for the remaining f -block lanthanide and actinide X-O pairs (Tm³⁺-Yb³⁺ and Th⁴⁺-U⁶⁺).

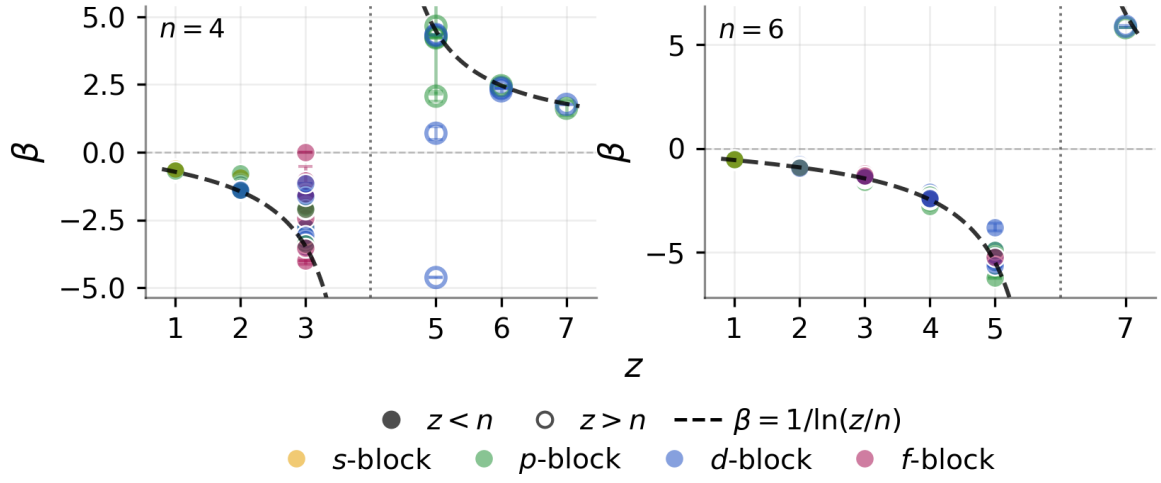


FIG. 16. Fitted slope β versus formal charge z for the individual cation–oxygen species summarized in main-text Fig. 1, at $n = 4$ (left) and $n = 6$ (right). Filled markers show species with $z < n$ (negative β), open markers species with $z > n$ (positive β), and dashed curves the parameter-free prediction $\beta = 1/\ln(z/n)$. Markers are colored by cation block and drawn semi-transparent, so darker points mark loci where the fitted slopes of multiple species coincide.

PER-SPECIES SLOPES AT FIXED COORDINATION

Figure 16 resolves the box-and-whisker summary of main-text Fig. 1 into the individual cation–oxygen species from which the coordination-number-resolved fits above were obtained.

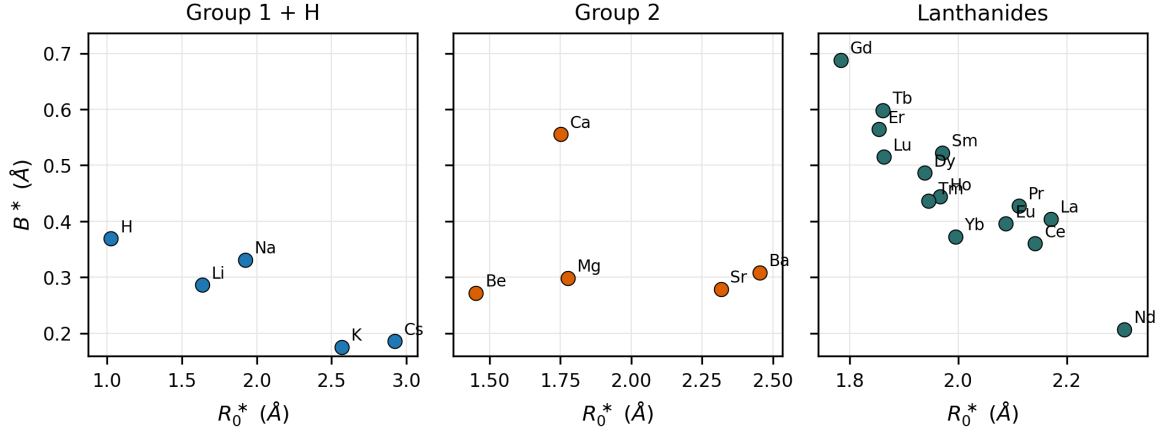


FIG. 17. Characteristic (R_0^*, B^*) for Group 1 including H^+ (left), Group 2 (center), and trivalent lanthanides (right).

CHARACTERISTIC PAIRS AND BOND-LENGTH IDENTITY

Figure 17 shows the characteristic pairs (R_0^*, B^*) for Group 1 (including H^+), Group 2, and the trivalent lanthanides.

For each species and coordination shell n , the fitted points define a line $B = \beta^{(n)}R_0 + \beta_0^{(n)}$. The characteristic pair is the weighted least-spread intersection of those n -resolved lines. If the final line fit for shell n contains N_n structures, we assign weight $w_n = N_n$ and define the weighted means

$$\bar{\beta} = \frac{\sum_n w_n \beta^{(n)}}{\sum_n w_n}, \quad \bar{\beta}_0 = \frac{\sum_n w_n \beta_0^{(n)}}{\sum_n w_n}.$$

The characteristic abscissa is then

$$R_0^* = -\frac{\sum_n w_n (\beta^{(n)} - \bar{\beta})(\beta_0^{(n)} - \bar{\beta}_0)}{\sum_n w_n (\beta^{(n)} - \bar{\beta})^2},$$

which is the point that minimizes the weighted variance of the line-by-line values $B_n(R_0) = \beta^{(n)}R_0 + \beta_0^{(n)}$. The characteristic ordinate and its spread are

$$B^* = \frac{\sum_n w_n B_n(R_0^*)}{\sum_n w_n}, \quad \sigma_B = \left[\frac{\sum_n w_n (B_n(R_0^*) - B^*)^2}{\sum_n w_n} \right]^{1/2}.$$

The tabulated n_{CN} counts how many distinct coordination shells contribute to that intersection, while the weights entering the intersection are the fitted structure counts in those shells.

Figure 18 validates the bond-length identity $R_0 \approx \bar{R} + B \ln(z/n)$ across 44,844 screening-branch cation-oxygen structures (records with $B > 0$, $0 < R_0 < 4$ Å, and $z \neq n$), with Pearson $r = 0.976$ and MAE = 0.019 Å.

s-BLOCK SCREENING-CENTROID CONTROLS

For the ten Group 1/Group 2 ground-state binary oxides used in Fig. 2 of the main text, the DFT screening centroid r_{eff} is compared against three linear descriptors: the bond-valence shell centroid m_i , the fitted bond-valence intercept R_0 , and the Shannon cation radius. Leave-one-out mean absolute errors are 0.006 Å for m_i , 0.078 Å for R_0 , and 0.095 Å for the Shannon-radius control, showing that the $r_{\text{eff}} - m_i$ collapse is substantially sharper than a generic monotone size trend.

Each oxide contributes one point (m_i, r_{eff}) to Fig. 2. For a given oxide, the bond-valence shell centroid is computed from the same fitted (R_0, B) and coordination number n used for its bond-valence record, taking the n shortest M-O distances in the ground-state structure and forming

$$p_{ij} = \frac{e^{-R_{ij}/B}}{\sum_{k=1}^n e^{-R_{ik}/B}}, \quad m_i = \sum_{j=1}^n p_{ij} R_{ij}.$$

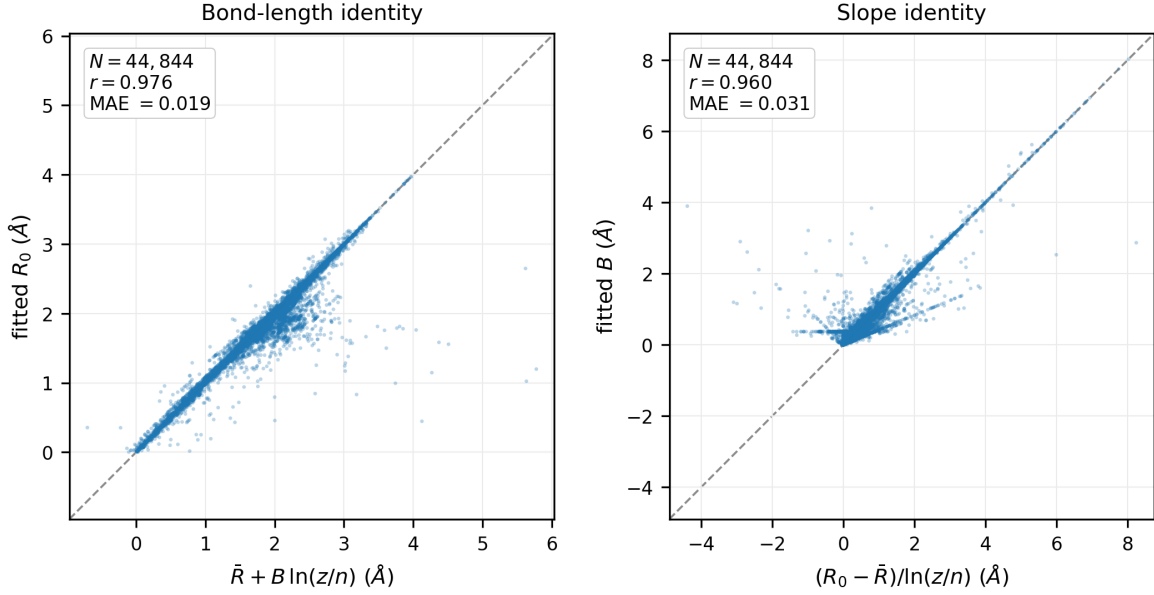


FIG. 18. Left: parity between fitted R_0 and $\bar{R} + B \ln(z/n)$ for 44,844 screening-branch cation–oxygen structures. Right: parity between fitted B and $(R_0 - \bar{R})/\ln(z/n)$ for the same set.

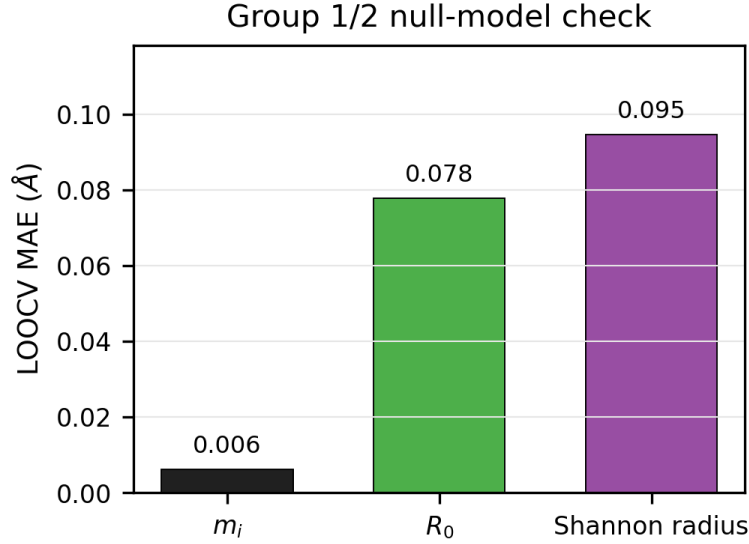


FIG. 19. Leave-one-out null-model check for the ten Group 1/Group 2 binary oxides used in Fig. 2 of the main text. Bars report the mean absolute error of linear regressions against the bond-valence shell centroid m_i , the fitted intercept R_0 , and the Shannon cation radius.

The DFT-side centroid is then built bond by bond over all oxygen neighbors with $R_{ij} \leq 3.5$ Å. Each such bond contributes one oxygen-side path centroid $r_{\text{eff}}^{(\text{path})}$, and the oxide-level r_{eff} plotted in Fig. 2 is the simple arithmetic mean of those finite bond-path centroids.

The offset in the main-text regression is quantified directly from that same discrete bond-path construction. Defining the oxygen-side pre-cusp valence centroid shift by

$$\Delta_{\text{O, val}} := \langle R_{ij} - r_{\text{eff}}^{(\text{path})} \rangle,$$

let $\{r_g, \rho_g\}_{g=1}^{200}$ denote the total-density samples on the nearest-image straight M–O segment at 200 uniformly spaced points. The bond critical-point proxy is the interior density minimum on the interval $0.15R_{ij} < r < 0.85R_{ij}$. Denote its location and value by r_{BCP} and ρ_{BCP} . The oxygen-side pre-cusp window is the subset of sampled points with $r_{\text{BCP}} \leq r_g \leq 0.93R_{ij}$, finite ρ_g , and positive finite Thomas–Fermi screening length $\lambda_{\text{TF}}(\rho_g)$. The Thomas–Fermi

length enters here only as a positivity filter on the sample window and does not appear in the centroid weights or in any of the regression results. The `thomas_fermi` tag in the Fig. 2 file label reflects only the screening interpretation of r_{eff} , not its construction. On that discrete window we use the excess-density weights $w_g = \max[\rho_g - \rho_{\text{BCP}}, 0]$. In the rare case that these clipped weights vanish identically, we use $w_g = \rho_g$ instead. Thus

$$r_{\text{eff}}^{(\text{path})} = \frac{\sum_g w_g r_g}{\sum_g w_g},$$

and the oxide-level centroid used in Fig. 2 is the arithmetic mean of $r_{\text{eff}}^{(\text{path})}$ over those first-shell M–O bonds. No spline fitting, path relaxation, or topological bond-path search is introduced beyond this fixed straight-segment sampling rule. With that definition, the ten-oxide s -block shift fit is $\Delta_{\text{O, val}} = 0.0613 m_i + 0.2394 \text{ \AA}$ ($R^2 = 0.963$), with mean $0.3910 \pm 0.0269 \text{ \AA}$. Equivalently, $r_{\text{eff}} = 0.9387 m_i - 0.2394 \text{ \AA}$, whose rounded form is the main-text relation. The intercept is therefore the zero-radius limit of a directly measured oxygen-side screening shift, not an unconstrained empirical parameter.

Species-level screening length from the characteristic pair. The shell-level Yukawa inversion in the main text is written in terms of the representative radius R^* , chosen operationally as the Boltzmann shell centroid m_i . To pass from that shell-level relation to a species-level constant, we evaluate the inversion at the characteristic fixed point (R_0^*, B^*) on the characteristic reference shell defined by $m_i = R_0^*$. Using the exact identity

$$R_0 = m_i + B(\ln q_i - H_i),$$

this reference-shell condition is equivalent at the fixed point to $H_i = \ln q_i$, so the representative radius entering the Yukawa inversion is $R^* = R_0^*$. Substituting $B = B^*$ and $R^* = R_0^*$ into the positive screening branch therefore gives

$$\lambda^* = \frac{-R_0^* + \sqrt{R_0^{*2} + 4B^*R_0^*}}{2},$$

which is the species-level screening length reported in Table I. For the two anti-screening species ($B^* < 0$) the same branch-continuous root is real whenever $B^* \geq -R_0^*/4$, a condition both satisfy. Their negative λ^* values are tabulated as signed Yukawa-branch diagnostics rather than physical screening lengths. The second root of the same quadratic is tabulated as λ_-^* for every species. On the anti-screening branch it also lies in the admissible interval $(-R_0^*, 0)$. On the screening branch it lies below $-R_0^*$, outside the admissible interval, and is reported for completeness.

Crystal-chemical anchoring of the second root. Because the two roots of the inversion quadratic sum to $-R_0^*$, the magnitude of the second root on the screening branch is $|\lambda_-^*| = R_0^* + \lambda^*$, the characteristic shell radius extended by one screening length. Figure 20 compares this quantity with an independent crystal-chemical scale. Each species is assigned a population-weighted Shannon crystal radius, obtained by interpolating the Shannon crystal radii [2] linearly in coordination number and averaging over the coordination numbers observed for that species in the fitted corpus, weighted by fitted structure counts. The high-spin radius is used where both spin states are tabulated. Unit-slope fits $|\lambda_-^*| = r_{\text{cr}} + c(z)$ describe each formal-charge class with 0.05–0.13 Å rms scatter, and the fitted offsets follow $c(z) = 1.05 + 0.092z \text{ \AA}$ ($r = 0.97$, inset). The constant term is an oxygen-radius-scale length, and the increment raises the effective cation–oxygen contact distance by about 0.09 Å per unit of formal charge. The figure shows the 96 cationic screening-branch species with tabulated Shannon radii. H^+ , P^{4+} , Co^+ , and Sn^{2+} lack entries at any coordination number, and Cl^- and the two anti-screening species are excluded.

SCREENING LENGTHS FOR THE ANALYZED SPECIES

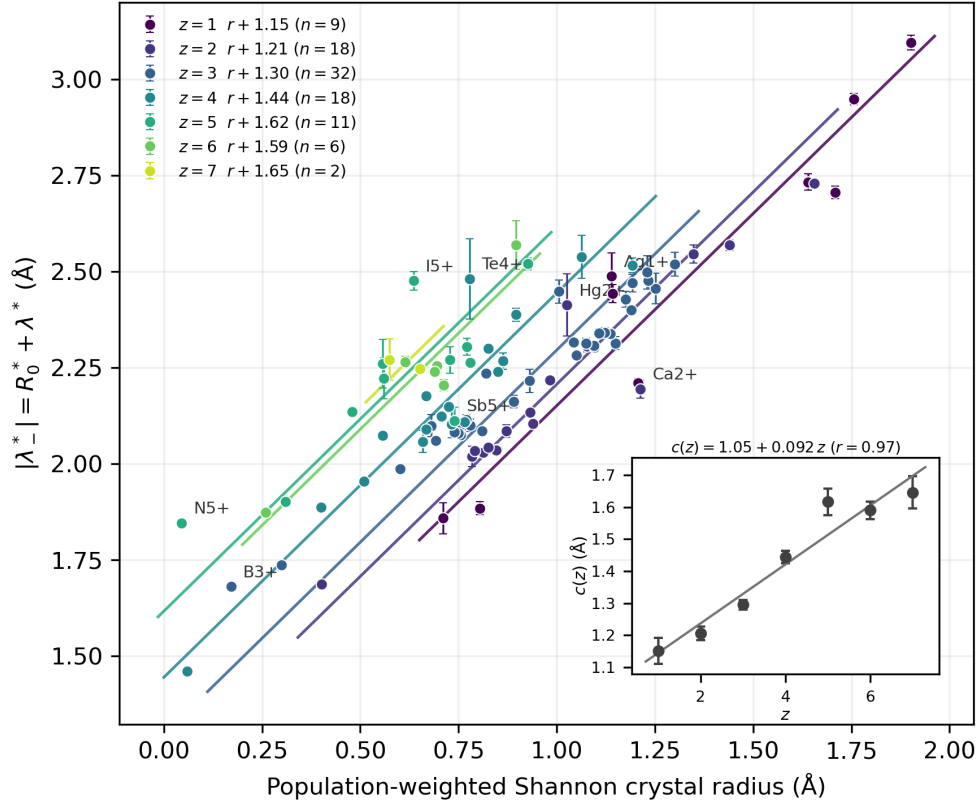


FIG. 20. Magnitude of the second inversion root, $|\lambda^*| = R_0^* + \lambda^*$, versus the population-weighted Shannon crystal radius [2] for 96 screening-branch cation species. Lines are unit-slope fits $|\lambda^*| = r_{cr} + c(z)$ for each formal charge z . Error bars are the propagated σ_λ of Table I. The inset shows the fitted offsets $c(z)$ with standard errors and the linear law $c(z) = 1.05 + 0.092 z$ ($r = 0.97$). Labels mark the largest per-class residuals.

TABLE I: Characteristic pairs (R_0^*, B^*) for all 103 cation–oxygen species and Yukawa screening lengths λ^* , ordered by atomic number. All values are in Å. λ^* is computed from the branch-continuous inversion $\lambda^* = (-R_0^* + \sqrt{R_0^{*2} + 4B^*R_0^*})/2$ in the main text, which is real for $B^* \geq -R_0^*/4$. For the 101 screening-branch species ($B^* > 0$) it is positive and physical. For the two anti-screening species (Mo^{3+} and Sb^{3+} , $B^* < 0$) the same root is negative and is tabulated as a signed Yukawa-branch diagnostic rather than a physical screening length, because the first-order expansion does not adequately describe their bonding. The second root of the same quadratic, $\lambda_-^* = (-R_0^* - \sqrt{R_0^{*2} + 4B^*R_0^*})/2$, is listed for every species in the λ_-^* column. On the anti-screening branch it also lies in the admissible interval $(-R_0^*, 0)$. On the screening branch it lies below $-R_0^*$, outside the admissible interval, and is reported for completeness. n_{CN} is the number of distinct coordination shells used to determine (R_0^*, B^*) . Parenthetical uncertainties on B^* are the weighted standard deviation σ_B of the n -resolved lines at R_0^* . Uncertainties on λ^* are propagated via $\sigma_\lambda = |\partial\lambda^*/\partial B^*| \sigma_B$. Entries with $n_{\text{CN}} = 2$ have $\sigma_B = 0$ by construction and should be regarded as provisional.

Species	Block	R_0^*	B^*	λ^*	λ_-^*	n_{CN}
H^{1+}	s	1.026	0.369(9)	0.288(6)	-1.314(6)	9
Li^{1+}	s	1.636	0.286(22)	0.248(17)	-1.884(17)	10
Be^{2+}	s	1.452	0.272(0)	0.234(0)	-1.686(0)	2
B^{3+}	p	1.376	0.372(9)	0.304(6)	-1.681(6)	7
C^{4+}	p	1.337	0.135(4)	0.123(3)	-1.460(3)	3

Continued on next page

TABLE I – continued from previous page

Species	Block	R_0^*	B^*	λ^*	λ_-^*	n_{CN}
N^{3+}	p	1.431	0.370(0)	0.305(0)	-1.736(0)	2
N^{5+}	p	1.492	0.437(0)	0.354(0)	-1.845(0)	2
Na^{1+}	s	1.923	0.331(15)	0.288(11)	-2.211(11)	10
Mg^{2+}	s	1.776	0.298(16)	0.260(12)	-2.036(12)	5
Al^{3+}	p	1.648	0.407(9)	0.338(6)	-1.985(6)	6
Si^{4+}	p	1.652	0.269(9)	0.235(7)	-1.887(7)	5
P^{4+}	p	1.552	0.004(0)	0.004(0)	-1.556(0)	2
P^{5+}	p	1.625	0.324(7)	0.277(6)	-1.902(6)	4
S^{4+}	p	1.644	0.370(0)	0.311(0)	-1.955(0)	3
S^{6+}	p	1.610	0.307(0)	0.264(0)	-1.874(0)	2
Cl^{1-}	p	1.804	0.370(0)	0.315(0)	-2.119(0)	2
K^{1+}	s	2.568	0.175(24)	0.165(21)	-2.733(21)	12
Ca^{2+}	s	1.751	0.555(35)	0.443(24)	-2.195(24)	8
Sc^{3+}	d	1.961	0.222(19)	0.201(16)	-2.162(16)	6
Ti^{3+}	d	2.009	0.080(0)	0.077(0)	-2.086(0)	2
Ti^{4+}	d	1.881	0.248(43)	0.222(35)	-2.103(35)	6
V^{3+}	d	1.782	0.375(22)	0.318(16)	-2.100(16)	6
V^{4+}	d	1.821	0.353(15)	0.303(11)	-2.124(11)	5
V^{5+}	d	1.849	0.448(75)	0.373(53)	-2.221(53)	6
Cr^{2+}	d	1.744	0.435(14)	0.360(10)	-2.104(10)	3
Cr^{3+}	d	1.761	0.370(0)	0.314(0)	-2.075(0)	3
Cr^{4+}	d	1.665	0.670(0)	0.512(0)	-2.178(0)	2
Cr^{5+}	d	1.821	0.544(93)	0.439(63)	-2.260(63)	3
Mn^{2+}	d	1.830	0.354(6)	0.304(5)	-2.134(5)	7
Mn^{3+}	d	1.810	0.357(18)	0.305(13)	-2.116(13)	7
Mn^{4+}	d	1.830	0.296(0)	0.260(0)	-2.090(0)	2
Fe^{2+}	d	1.760	0.385(22)	0.325(16)	-2.085(16)	6
Fe^{3+}	d	1.799	0.352(9)	0.302(6)	-2.101(6)	5
Fe^{4+}	d	1.816	0.394(0)	0.333(0)	-2.149(0)	2
Co^{1+}	d	1.584	0.370(0)	0.310(0)	-1.893(0)	2
Co^{2+}	d	1.712	0.384(6)	0.323(5)	-2.035(5)	4
Co^{3+}	d	1.776	0.354(5)	0.303(3)	-2.078(3)	4
Co^{4+}	d	1.774	0.330(38)	0.284(29)	-2.058(29)	3
Ni^{2+}	d	1.766	0.304(8)	0.264(6)	-2.030(6)	5
Ni^{3+}	d	1.773	0.364(12)	0.310(9)	-2.083(9)	3
Cu^{1+}	d	1.585	0.319(53)	0.273(40)	-1.858(40)	4
Cu^{2+}	d	1.715	0.358(36)	0.304(27)	-2.019(27)	7
Cu^{3+}	d	1.708	0.481(42)	0.391(29)	-2.099(29)	3
Zn^{2+}	d	1.760	0.328(7)	0.283(5)	-2.043(5)	4
Ga^{3+}	p	1.756	0.357(5)	0.304(4)	-2.060(4)	4
Ge^{4+}	p	1.771	0.354(15)	0.302(11)	-2.073(11)	4
As^{5+}	p	1.812	0.382(1)	0.324(1)	-2.136(1)	3
Rb^{1+}	s	2.809	0.146(16)	0.140(15)	-2.949(15)	10
Sr^{2+}	s	2.317	0.279(17)	0.252(14)	-2.568(14)	9
Y^{3+}	d	1.806	0.634(18)	0.497(12)	-2.303(12)	9
Zr^{4+}	d	1.979	0.330(27)	0.288(21)	-2.268(21)	6
Nb^{5+}	d	1.945	0.426(30)	0.359(22)	-2.304(22)	7
Mo^{3+}	d	2.324	-0.237(0)	-0.268(0)	-2.057(0)	2
Mo^{5+}	d	1.919	0.415(46)	0.351(34)	-2.270(34)	4
Mo^{6+}	d	1.948	0.368(19)	0.317(14)	-2.265(14)	5
Pd^{2+}	d	1.402	0.918(0)	0.632(0)	-2.035(0)	2
Ag^{1+}	d	2.477	0.011(61)	0.011(60)	-2.489(60)	8
Ag^{2+}	d	1.778	0.548(0)	0.439(0)	-2.217(0)	2
Cd^{2+}	d	2.083	0.247(18)	0.223(15)	-2.306(15)	6
In^{3+}	p	2.065	0.162(34)	0.151(30)	-2.216(30)	5
Sn^{2+}	p	1.903	0.607(45)	0.484(30)	-2.387(30)	4
Sn^{4+}	p	1.845	0.567(9)	0.455(6)	-2.300(6)	3
Sb^{3+}	p	2.434	-0.341(87)	-0.410(131)	-2.025(131)	5
Sb^{5+}	p	1.990	0.130(39)	0.122(35)	-2.112(35)	4

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TABLE I – continued from previous page

Species	Block	R_0^*	B^*	λ^*	λ_-^*	n_{CN}
Te ⁴⁺	<i>p</i>	2.023	0.562(151)	0.458(104)	-2.481(104)	7
Te ⁶⁺	<i>p</i>	1.937	0.370(0)	0.318(0)	-2.254(0)	2
I ⁵⁺	<i>p</i>	2.077	0.476(34)	0.399(24)	-2.476(24)	6
I ⁷⁺	<i>p</i>	1.964	0.324(4)	0.283(3)	-2.247(3)	3
Cs ¹⁺	<i>s</i>	2.921	0.185(21)	0.175(19)	-3.096(19)	11
Ba ²⁺	<i>s</i>	2.453	0.308(11)	0.277(9)	-2.730(9)	10
La ³⁺	<i>f</i>	2.171	0.404(41)	0.348(31)	-2.519(31)	12
Ce ³⁺	<i>f</i>	2.142	0.360(50)	0.314(39)	-2.456(39)	9
Pr ³⁺	<i>f</i>	2.112	0.427(30)	0.364(22)	-2.476(22)	7
Nd ³⁺	<i>f</i>	2.307	0.207(50)	0.191(43)	-2.498(43)	8
Sm ³⁺	<i>f</i>	1.971	0.522(13)	0.429(9)	-2.400(9)	8
Eu ³⁺	<i>f</i>	2.088	0.396(28)	0.340(21)	-2.428(21)	6
Gd ³⁺	<i>f</i>	1.784	0.688(28)	0.530(17)	-2.314(17)	6
Tb ³⁺	<i>f</i>	1.862	0.598(18)	0.476(12)	-2.338(12)	5
Dy ³⁺	<i>f</i>	1.938	0.486(18)	0.403(13)	-2.341(13)	5
Ho ³⁺	<i>f</i>	1.967	0.444(16)	0.373(11)	-2.340(11)	5
Er ³⁺	<i>f</i>	1.854	0.564(10)	0.453(7)	-2.307(7)	5
Tm ³⁺	<i>f</i>	1.946	0.436(18)	0.367(13)	-2.313(13)	4
Yb ³⁺	<i>f</i>	1.995	0.372(7)	0.321(5)	-2.316(5)	4
Lu ³⁺	<i>f</i>	1.863	0.515(13)	0.420(9)	-2.283(9)	5
Hf ⁴⁺	<i>d</i>	1.929	0.362(16)	0.311(12)	-2.240(12)	4
Ta ⁵⁺	<i>d</i>	1.924	0.400(14)	0.340(10)	-2.265(10)	5
W ⁶⁺	<i>d</i>	1.936	0.305(20)	0.268(15)	-2.205(15)	4
Re ⁶⁺	<i>d</i>	1.928	0.363(0)	0.312(0)	-2.241(0)	2
Re ⁷⁺	<i>d</i>	1.948	0.377(72)	0.323(54)	-2.271(54)	4
Ir ⁴⁺	<i>d</i>	1.984	0.133(0)	0.125(0)	-2.109(0)	2
Pt ²⁺	<i>d</i>	1.785	0.370(0)	0.315(0)	-2.100(0)	2
Au ³⁺	<i>d</i>	1.924	0.363(13)	0.312(10)	-2.237(10)	3
Hg ¹⁺	<i>d</i>	2.230	0.233(28)	0.213(24)	-2.443(24)	4
Hg ²⁺	<i>d</i>	1.929	0.605(122)	0.484(81)	-2.413(81)	6
Tl ¹⁺	<i>p</i>	2.459	0.272(19)	0.247(16)	-2.705(16)	10
Tl ³⁺	<i>p</i>	2.054	0.469(42)	0.394(30)	-2.448(30)	4
Pb ²⁺	<i>p</i>	2.240	0.348(31)	0.306(24)	-2.547(24)	10
Pb ⁴⁺	<i>p</i>	2.080	0.353(23)	0.308(17)	-2.387(17)	3
Bi ³⁺	<i>p</i>	2.104	0.430(31)	0.366(23)	-2.470(23)	8
Th ⁴⁺	<i>f</i>	2.128	0.459(26)	0.388(19)	-2.516(19)	5
U ⁴⁺	<i>f</i>	1.938	0.787(91)	0.600(56)	-2.539(56)	3
U ⁵⁺	<i>f</i>	2.089	0.521(23)	0.431(16)	-2.521(16)	3
U ⁶⁺	<i>f</i>	2.080	0.603(92)	0.489(63)	-2.569(63)	7

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